



Effect of ambient relative humidity and particulate matter on the levels of asthma therapy

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ABSTRACT

Objective: We aimed to investigate the associations of the daily mean and short-term variation in ambient relative humidity (RH), temperature, and fine particulate matter (PM_{2.5}) with asthma treatment steps.

Methods: We consecutively recruited 288 newly diagnosed adult patients with asthma and followed them for at least 6 months according to the Global Initiative for Asthma (GINA) strategy. Odds ratios (ORs) for higher GINA treatment steps required to achieve asthma control were estimated using logistic regression. We also conducted exposure-response analyses to explore potentially nonlinear relationships between environmental exposure and asthma treatment step across seasons.

Results: A 1% increase in the 1-day RH difference and a 1-μg/m³ increase in the 7-day PM_{2.5} difference were associated with 1.054-fold and 1.052-fold higher odds of step 4 treatment, respectively. A 1% increase in the 1-day RH difference and a 1-μg/m³ increase in the 7-day PM_{2.5} difference were associated with 1.047-fold and 1.056-fold higher odds of medium-dose inhaled corticosteroids plus long-acting beta-agonists (ICS + LABA), respectively. A 1% increase in the 7-day RH mean and a 1-μg/m³ increase in the 30-day PM_{2.5} difference were associated with 1.075-fold and 1.205-fold higher odds of triple inhaler therapy and biologics use, respectively. Exposure-response analyses suggested noticeable seasonal patterns for temperature and RH in relation to asthma treatment strategy.

Conclusions: Our exploratory and hypothesis-generating findings showed that greater short-term variation in RH and PM_{2.5} was associated with higher treatment intensity in adult asthma, particularly the use of medium-dose ICS + LABA and biologics. Higher RH mean was also associated with triple inhaler therapy. These associations were more evident during the cold season. Multicenter validation studies in larger population studies are warranted.

Keywords: Asthma, Air pollution, Humidity, Particulate matter, Environmental exposure

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INTRODUCTION

Asthma is a chronic respiratory disease characterized by variable respiratory symptoms and variable expiratory airflow over time. Asthma severity is defined retrospectively by the level of treatment required to achieve symptom control and prevent exacerbations, a subset of patients progress to severe asthma requiring high-intensity treatment.¹ Environmental exposures can influence both asthma symptoms and treatment response. The Intergovernmental Panel on Climate Change (IPCC) has highlighted the importance of meteorological conditions and air pollution in respiratory morbidity and mortality.²

Poor air quality, extreme temperatures, and fluctuations in humidity may aggravate asthma symptoms and increase the risk of exacerbations.^{3,4} Fine particulate matter with an aerodynamic diameter of $<2.5\ \mu\text{m}$ ($\text{PM}_{2.5}$) has also been linked to asthma incidence, impaired lung function, and poorer symptom control.⁵⁻⁷ Previous studies have reported that exposure to $\text{PM}_{2.5}$ is associated with increased asthma medication use.^{8,9} Together, these findings suggest that meteorological factors and particulate air pollution may contribute to greater treatment difficulty in patients with asthma.

Seasonal variation may further modify these associations. Changes in temperature, RH, allergen levels, and pollutant concentrations can affect asthma symptoms and treatment response across seasons.^{10,11} For example, pollen peaks in spring and summer may trigger allergic reactions and asthma exacerbations, whereas indoor heating and dry air during winter may irritate the airways.¹² Seasonal increases in PM exposure, including wildfire smoke episodes, may also worsen asthma symptoms and alter treatment needs.^{13,14} These observations suggest that meteorological conditions and $\text{PM}_{2.5}$ may influence asthma treatment strategies differently across seasons.

Based on this background, we hypothesized that ambient RH, temperature, and $\text{PM}_{2.5}$ would

be associated with the treatment step required to achieve asthma control, and that these associations might vary by season. Therefore, the objective of this study was to investigate the associations of the daily mean and short-term variation in ambient RH, temperature, and $\text{PM}_{2.5}$ with asthma treatment strategy in adults with newly diagnosed asthma.

METHODS

Study population

A prospective observational study was conducted in 288 adult patients with asthma who were consecutively recruited from the outpatient pulmonary clinic of a medical center in Taoyuan City, Taiwan, between March 2019 and December 2020 and followed for at least 6 months, with the GINA treatment step at 6 months designated as the primary study outcome. Eligible participants were aged >18 years and were newly diagnosed with asthma by a pulmonologist according to the Global Initiative for Asthma (GINA) based on clinical history, physical examination, and lung function assessment.¹ Patients attended scheduled follow-up visits, and the pulmonologist together with a trained educator provided in-person education regarding asthma control, self-management planning, and correct inhaler technique. Only patients who were followed regularly for at least 6 months were included in the analysis. Baseline data included age, sex, body mass index (BMI), smoking status (current smoker, ex-smoker, or never smoker), family history of asthma, comorbidities, and residential location during the previous 6 months. IgE level, eosinophil count, lung function parameters, and eosinophil cationic protein level were recorded. We excluded patients who could not be followed regularly for 6 months or who had confirmed chronic obstructive pulmonary disease, bronchiectasis, malignancy, or chronic inflammatory conditions unrelated to asthma. No a priori sample size calculation was performed because this analysis was based on consecutively followed patients who met the study criteria during the study period. All patients provided

written informed consent. The study protocol was approved by the Chang Gung Medical Foundation Institutional Review Board (No. 201900211B0).

Asthma treatment and health outcome measurement

The GINA asthma treatment strategy includes 5 treatment steps designed to match treatment intensity to the level required to achieve asthma control.¹ For treatment classification in this study, we used the GINA 2018 strategy throughout, because the protocol was developed in late 2018 and this provided a uniform framework before GINA 2019 introduced alternative reliever pathways. In this study, step 1 was defined as short-acting beta-agonist as needed; step 2 as inhaled corticosteroids (ICS); step 3 as low-dose ICS plus long-acting beta-agonist (LABA); step 4 as medium- or high-dose ICS + LABA; and step 5 as triple inhaler therapy (ICS + LABA + long-acting muscarinic antagonist), any inhaler regimen combined with maintenance oral corticosteroids, or any inhaler regimen combined with biologics. The GINA treatment step recorded at 6 months after recruitment was used as the main study outcome.

Ambient RH, temperature, and PM_{2.5} assessment

Meteorological and PM_{2.5} data from 2018 to 2020 were collected for analysis. Daily RH and temperature data were obtained from Central Weather Bureau (CWB) stations, and hourly PM_{2.5} concentration data were obtained from the Environmental Protection Administration (EPA). Hourly PM_{2.5} data were aggregated into daily averages. We then estimated the spatial distribution of temperature, RH, and PM_{2.5} across Taiwan from 2018 to 2020 using radial basis function (RBF) interpolation. For each participant, RH, temperature, and PM_{2.5} mean values and difference values were calculated for 1-, 7-, and 30-day periods before the index date. The mean value was calculated from day t-1 to the day of the index date, where t represented 1, 7, or 30 days. The difference value was defined as the current mean minus the mean from the preceding period of the same duration. The application of RBF interpolation in our group has been described previously,

and additional methodological details are provided in the Supplementary Materials.¹⁵

Statistical analyses

Continuous variables are presented as mean $\pm$ standard deviation (SD), and categorical variables are presented as counts and percentages. Winsorization was applied to values outside the 1st and 99th percentiles to reduce the influence of extreme outliers.¹⁶ Normality testing was performed before analysis. Correlations among 1-, 7-, and 30-day means and differences in RH, temperature, and PM_{2.5} were examined using Pearson correlation coefficients. The primary analyses used logistic regression to estimate the ORs of higher asthma treatment steps in relation to environmental exposures. Age, sex, BMI, and smoking status were included as covariates. To explore potential nonlinear exposure-response relationships, we additionally used generalized additive models with smoothing splines (4 degrees of freedom) and examined the curves separately for the cold season (November–April) and warm season (May–October). Regression analyses were performed using SPSS for Windows version 20.0 (SPSS Inc., Chicago, IL, USA), and exposure-response analyses were conducted in R using the mgcv package (version 4.2.2). A 2-sided P value < 0.05 was considered statistically significant.

Patient and public involvement

Patients and the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

RESULTS

Characteristics of study subjects

Table 1 summarizes the characteristics of the 288 patients included in this study. Most patients were male (52.8%), and the mean age was 61.9 years. Regarding smoking status, 14.2% were current smokers, 28.1% were ex-smokers, and 57.6% were never smokers. The mean BMI was 25.49 kg/m². Family history of asthma was present in 30.2% of patients; 66.7% had allergic rhinitis, 25.3% had rhinosinusitis, and 9.4% had aspirin sensitivity. Mean IgE level, eosinophil count, and eosinophil

Variables	Total
N	288
Age (years), mean $\pm$ SD	61.9 $\pm$ 14.0
Male, n (%)	152 (52.8)
BMI (kg/m ²), mean $\pm$ SD	25.49 $\pm$ 4.0
Smoking status, n (%)	
Current smoker	41 (14.2)
Ex-smoker	81 (28.1)
Non-smoker	166 (57.6)
Family history of asthma, n (%)	87 (30.2)
Allergic rhinitis, n (%)	192 (66.7)
Rhinosinusitis, n (%)	73 (25.3)
Aspirin sensitivity, n (%)	27 (9.4)
IgE level (KU/L), mean $\pm$ SD	170.1 $\pm$ 356.7
Eosinophil count, mean $\pm$ SD	137.7 $\pm$ 158.6
Eosinophil cationic protein (μ g/L), mean $\pm$ SD	3.1 $\pm$ 7.0
Lung function, mean $\pm$ SD	
FEV ₁ , %	59.5 $\pm$ 19.3
FVC, %	81.9 $\pm$ 20.3
FEV1/FVC	0.73 $\pm$ 0.10
GINA treatment strategy, n (%)	
Step 2	29 (10.1)
Step 3	88 (30.6)
Step 4	96 (33.3)
Step 5	75 (26.0)
Environmental exposure	
RH, % (min-max)	
1-Day average	76.5 (51.4-97.5)
7-Day average	76.2 (64.3-89.8)
30-Day average	75.7 (67.1-86.9)
1-Day difference	-0.6 (-26.0-27.6)
7-Day difference	0.7 (-22.4-14.9)
30-Day difference	-0.2 (-12.2-9.7)
Temperature, °C (min-max)	
1-Day average	24.4 (14.1-31.2)
7-Day average	24.6 (16.1-31.4)
30-Day average	13.9 (6.3-23.6)
1-Day difference	-0.02 (-4.9-4.0)
7-Day difference	-0.2 (-5.0-5.6)
30-Day difference	-0.07 (-4.7-6.2)
PM _{2.5} , μ g/m ³ (min-max)	
1-Day average	13.9 (3.7-45.2)
7-Day average	13.4 (5.0-27.2)

(continued)

Variables	Total
30-Day average	13.9 (6.3–23.6)
1-Day difference	0.8 (–22.1–34.2)
7-Day difference	–0.4 (–16.7–15.9)
30-Day difference	0.06 (–8.4–6.6)

Table 1. (Continued) Basic characteristics in the study subjects. BMI, body mass index; FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity; GINA, Global Initiative for Asthma Strategy; RH, relative humidity; PM_{2.5}, particulate matter of 2.5 μ m in aerodynamic diameter

cationic protein level were 170.1 KU/L, 137.7 cells/ μ L, and 3.1 μ g/L, respectively. Mean FEV₁ and FVC (% predicted), and FEV₁/FVC (absolute value) were 59.5% $\pm$ 19.3%, 81.9% $\pm$ 20.3%, and 0.73 $\pm$ 0.10, respectively. Although the spirometric profile did not always show a classic fixed obstructive pattern, this is not inconsistent with asthma, because asthma is defined by variable airflow and clinical variability rather than a requirement for persistent FEV₁/FVC <0.7¹. Distributions of the daily mean and difference in ambient RH, temperature, and PM_{2.5} for 1, 7, and 30 days are shown in [Supplementary Fig. 1](#). The 1-, 7-, and 30-day averages of RH, temperature, and PM_{2.5} were 75.7%–76.5% (range, 51.4%–97.5%), 24.4–24.9 °C (range, 14.1–31.4 °C), and 13.4–13.9 μ g/m³ (range, 3.7–45.2 μ g/m³), respectively. The corresponding 1-, 7-, and 30-day differences ranged from –26.0% to 27.6% for RH, –6.2 °C–5.6 °C for temperature, and –22.1 to 34.2 μ g/m³ for PM_{2.5}. Correlations among these exposure metrics are presented in [Supplementary Table S1](#). Daily means of RH (1 and 7 days) and temperature (1, 7, and 30 days) were negatively correlated with PM_{2.5}.

Associations of ambient RH, temperature, and PM_{2.5} with asthma treatment steps

Associations of ambient RH, temperature, and PM_{2.5} with asthma treatment strategy are shown in [Table 2](#). A 1% increase in the 1-day RH difference was associated with a 1.054-fold increase in the odds of step 4 treatment (95% CI: 1.016–1.093, $P < 0.05$). A 1- μ g/m³ increase in the 7-day PM_{2.5} difference was associated with a 1.052-fold increase in the odds of step 4 treatment (95% CI: 1.002–1.105, $P < 0.05$). In contrast, a 1- μ g/m³ increase in the 7-day PM_{2.5} difference was associated with a 0.929-fold decrease in the odds of step 5 treatment (95% CI: 0.881–0.980, $P < 0.05$). Because this inverse association was isolated and biologically counterintuitive, it should be interpreted cautiously. No significant associations were

observed for mean RH, temperature, or PM_{2.5} with overall asthma treatment strategy.

Associations of ambient RH, temperature, and PM_{2.5} with step 4 treatment subgroups in asthma

Associations of ambient RH, temperature, and PM_{2.5} with the step 4 treatment subgroups are shown in [Table 3](#). A 1% increase in the 1-day RH difference was associated with a 1.047-fold increase in the odds of medium-dose ICS + LABA use (95% CI: 1.009–1.086, $P < 0.05$). A 1- μ g/m³ increase in the 7-day PM_{2.5} difference was associated with a 1.056-fold increase in the odds of medium-dose ICS + LABA use (95% CI: 1.004–1.112, $P < 0.05$). No significant associations were observed for mean RH, temperature, or PM_{2.5} with the step 4 treatment subgroups.

Associations of ambient RH, temperature, and PM_{2.5} with step 5 treatment subgroups in asthma

Associations of ambient RH, temperature, and PM_{2.5} with the step 5 treatment subgroups are shown in [Table 4](#). A 1% increase in the 7-day RH mean was associated with a 1.075-fold increase in the odds of triple inhaler therapy (95% CI: 1.015–1.137, $P < 0.05$). A 1- μ g/m³ increase in the 30-day PM_{2.5} difference was associated with a 1.205-fold increase in the odds of biologics use (95% CI: 1.011–1.436, $P < 0.05$).

Exposure-response relationships of ambient RH, temperature, and PM_{2.5} with asthma treatment steps

Exposure-response relationships between ambient means ([Fig. 1](#) and [Supplementary Figs. S2–S3](#)) and differences ([Supplementary Figs. S4–S6](#)) in RH, temperature, and PM_{2.5} and asthma treatment strategy during the warm and cold seasons are presented. Nonlinear exposure-response relationships were observed for RH, temperature, and PM_{2.5} in both their mean and

	Step 2	Step 3	Step 4	Step 5
	Or (95% CI)	Or (95% CI)	Or (95% CI)	Or (95% CI)
RH mean, %				
1-day	0.994 (0.949–1.040)	0.993 (0.963–1.024)	1.004 (0.975–1.034)	1.007 (0.976–1.040)
7-day	0.969 (0.901–1.042)	0.998 (0.952–1.047)	0.991 (0.946–1.038)	1.029 (0.979–1.082)
30-day	0.990 (0.890–1.101)	1.015 (0.947–1.088)	0.973 (0.908–1.042)	1.020 (0.947–1.098)
RH difference, %				
1-day	0.997 (0.962–1.034)	0.983 (0.950–1.018)	1.054 (1.016–1.093)	0.972 (0.937–1.008)
7-day	0.978 (0.922–1.039)	0.986 (0.946–1.029)	1.000 (0.959–1.042)	1.013 (0.968–1.060)
30-day	1.017 (0.939–1.101)	0.997 (0.940–1.058)	1.007 (0.950–1.068)	0.997 (0.936–1.062)
Temperature mean, °C				
1-day	0.991 (0.907–1.083)	1.011 (0.954–1.072)	0.971 (0.917–1.028)	1.024 (0.963–1.089)
7-day	1.021 (0.929–1.121)	1.011 (0.951–1.074)	0.961 (0.905–1.020)	1.023 (0.959–1.091)
30-day	1.003 (0.913–1.103)	1.008 (0.947–1.073)	0.971 (0.914–1.032)	1.023 (0.957–1.093)
Temperature difference, °C				
1-day	0.942 (0.733–1.212)	0.981 (0.830–1.160)	0.917 (0.779–1.080)	1.158 (0.965–1.389)
7-day	0.977 (0.778–1.228)	1.111 (0.959–1.287)	1.016 (0.879–1.173)	0.882 (0.751–1.035)
30-day	1.031 (0.890–1.195)	1.018 (0.923–1.123)	0.933 (0.846–1.028)	1.046 (0.945–1.158)
PM_{2.5} mean, µg/m³				
1-day	0.991 (0.939–1.046)	1.003 (0.970–1.038)	1.003 (0.970–1.037)	0.997 (0.962–1.034)
7-day	1.029 (0.947–1.119)	0.995 (0.940–1.052)	1.012 (0.958–1.069)	0.978 (0.922–1.039)
30-day	0.981 (0.873–1.103)	1.015 (0.941–1.094)	0.979 (0.909–1.055)	1.017 (0.939–1.101)

(continued)

	Step 2	Step 3	Step 4	Step 5
	Or (95% CI)	Or (95% CI)	Or (95% CI)	Or (95% CI)
PM_{2.5} difference, $\mu\text{g}/\text{m}^3$				
1-day	0.995 (0.941–1.053)	0.988 (0.951–1.025)	0.988 (0.953–1.025)	1.028 (0.990–1.067)
7-day	1.063 (0.984–1.147)	0.988 (0.941–1.037)	1.052 (1.002–1.105)	0.929 (0.881–0.980)
30-day	0.922 (0.816–1.041)	1.014 (0.935–1.101)	0.959 (0.886–1.039)	1.079 (0.988–1.179)

Table 2. (Continued) Associations of the daily mean and differences in the ambient particulate matter with an aerodynamic diameter of $<2.5 \mu\text{m}$ (PM_{2.5}), relative humidity (RH), and temperature for 1-, 7-, and 30-day means with the asthma treatment (step 2, step 3, step 4, and step 5). Data are presented as the odds ratio (OR) and 95% confident interval (CI). Abbreviations: OR, odds ratio; CI, confidence interval; particulate matter with an aerodynamic diameter of $<2.5 \mu\text{m}$ (PM_{2.5}); relative humidity (RH). Covariates adjusted for age, body mass index (BMI), and smoking status. The bold value indicates statistically significant ($p < 0.05$)

difference metrics. Seasonal patterning was most apparent for temperature and RH, with stronger effects on treatment strategy during the cold season. Similar seasonal patterns were observed for step 4 treatment subgroups and for triple inhaler therapy, oral corticosteroid use, and biologics use in step 5.

DISCUSSION

This study examined whether ambient meteorological conditions and PM_{2.5} were associated with the treatment intensity required to achieve asthma control. Several findings deserve emphasis. First, short-term variation in RH and PM_{2.5}, rather than their mean levels alone, was associated with higher treatment intensity. Specifically, higher 1-day RH difference and 7-day PM_{2.5} difference were associated with step 4 treatment, mainly medium-dose ICS + LABA use. Higher 7-day RH mean was associated with triple inhaler therapy, and higher 30-day PM_{2.5} difference was associated with biologics use. Second, exposure-response analyses suggested that the relationships of temperature and RH with treatment strategy varied by season, with more prominent associations during the cold season. Overall, these findings suggest that fluctuations in meteorological conditions and particulate air pollution may contribute to treatment difficulty in adult asthma.

We observed that mean PM_{2.5} and RH tended to decrease as mean temperature increased,

consistent with prior environmental studies showing that meteorological conditions can substantially affect PM_{2.5} concentration and humidity.¹⁷⁻²⁰ Higher temperature may promote atmospheric mixing and particle dispersion, thereby lowering particle mass concentration.^{19,20} In addition, warmer conditions may alter relative humidity through changes in moisture-holding capacity and atmospheric mixing rather than through any decrease in air volume.²¹ These observations support the concept that temperature may act as an upstream environmental modifier of both PM_{2.5} and RH exposure.

Short-term changes in meteorological factors were associated with more intensive asthma treatment in this study. Previous studies have shown that both high and low ambient RH and temperature may increase the risk of asthma exacerbation.^{22,23} Consistent with this general concept, we found that higher RH mean was associated with triple inhaler therapy and that greater RH difference was associated with step 4 treatment, particularly medium-dose ICS + LABA. Potential mechanisms include airway hyperresponsiveness, airway remodeling, and worsening respiratory symptoms after exposure to unfavorable humidity conditions. Experimental and clinical studies have shown that asthmatic airways are more susceptible than non-asthmatic airways to cold, dry air challenge, and that airway responsiveness may be

	Medium dose ICS-LABA Or (95% CI)	High dose ICS-LABA Or (95% CI)
RH mean, %		
1-day	1.002 (0.972–1.033)	1.009 (0.944–1.079)
7-day	0.993 (0.946–1.041)	0.992 (0.895–1.101)
30-day	0.983 (0.915–1.056)	0.939 (0.796–1.107)
RH difference, %		
1-day	1.047 (1.009–1.087)	1.049 (0.966–1.139)
7-day	0.999 (0.957–1.043)	1.003 (0.914–1.102)
30-day	1.009 (0.950–1.072)	0.995 (0.872–1.135)
Temperature mean, °C		
1-day	0.988 (0.932–1.048)	0.908 (0.797–1.034)
7-day	0.972 (0.914–1.034)	0.933 (0.815–1.067)
30-day	0.977 (0.918–1.040)	0.963 (0.842–1.102)
Temperature difference, °C		
1-day	0.931 (0.786–1.103)	0.900 (0.623–1.300)
7-day	1.048 (0.903–1.216)	0.853 (0.605–1.203)
30-day	0.967 (0.875–1.069)	0.800 (0.617–1.038)
PM_{2.5} mean, µg/m³		
1-day	1.010 (0.976–1.046)	0.960 (0.877–1.051)
7-day	1.011 (0.955–1.070)	1.005 (0.888–1.138)
30-day	0.967 (0.894–1.045)	1.055 (0.890–1.249)
PM_{2.5} difference, µg/m³		
1-day	0.991 (0.954–1.029)	0.984 (0.904–1.071)
7-day	1.056 (1.004–1.112)	0.998 (0.896–1.111)
30-day	0.962 (0.886–1.044)	0.977 (0.818–1.166)

Table 3. Associations of the daily mean and difference in the ambient particulate matter with an aerodynamic diameter of $<2.5 \mu\text{m}$ (PM_{2.5}), relative humidity (RH), and temperature for 1-, 7-, and 30-day averages with the step 4 treatments (medium dose ICS-LABA and high dose ICS-LABA). Data are presented as the odds ratio (OR) and 95% confident interval (CI). Abbreviations: CI, confidence interval; OR, ICS, inhaled corticosteroids; LABA, long-acting beta-agonist; odds ratio; particulate matter with an aerodynamic diameter of $<2.5 \mu\text{m}$ (PM_{2.5}); relative humidity (RH). Covariates adjusted for age, body mass index (BMI), and smoking status. The bold value indicates statistically significant ($p < 0.05$)

affected by the humidity of inspired air.^{24,25} Changes in ambient RH may also reduce expiratory flow and increase airway resistance.²⁶ Although our spirometric profile did not uniformly show persistent obstruction, this does not conflict with the diagnosis of asthma because

variable airflow rather than fixed airflow obstruction is the defining physiologic feature in GINA.¹ Further mechanistic studies are needed.

We also found that greater PM_{2.5} variation was associated with more intensive treatment, including step 4 therapy and biologics use.

	Triple inhaler (n = 48)	Chronic oral corticosteroids (n = 15)	Biologics (n = 12)
	Or (95% CI)	Or (95% CI)	Or (95% CI)
RH mean, %			
1-day	1.032 (0.995–1.070)	1.008 (0.948–1.073)	0.974 (0.918–1.033)
7-day	1.075 (1.015–1.139)	0.974 (0.882–1.074)	1.002 (0.922–1.089)
30-day	1.052 (0.970–1.142)	0.956 (0.822–1.112)	0.946 (0.847–1.056)
RH difference, %			
1-day	0.975 (0.937–1.015)	1.006 (0.936–1.081)	0.997 (0.930–1.069)
7-day	1.046 (0.992–1.103)	0.983 (0.902–1.070)	1.002 (0.922–1.089)
30-day	1.016 (0.946–1.092)	0.969 (0.857–1.096)	0.946 (0.847–1.056)
Temperature mean, °C			
1-day	0.991 (0.926–1.061)	1.082 (0.951–1.229)	1.034 (0.926–1.155)
7-day	0.996 (0.927–1.069)	1.093 (0.953–1.253)	1.011 (0.901–1.134)
30-day	0.995 (0.925–1.071)	1.103 (0.955–1.272)	1.027 (0.910–1.159)
Temperature difference, °C			
1-day	1.117 (0.912–1.366)	0.948 (0.670–1.341)	1.138 (0.821–1.576)
7-day	0.885 (0.739–1.059)	0.858 (0.623–1.181)	0.901 (0.681–1.194)
30-day	1.050 (0.938–1.176)	1.070 (0.884–1.298)	1.001 (0.831–1.206)
PM_{2.5} mean, µg/m³			
1-day	1.004 (0.965–1.044)	0.998 (0.930–1.071)	0.999 (0.935–1.067)
7-day	0.988 (0.925–1.055)	0.971 (0.862–1.095)	1.037 (0.935–1.151)
30-day	1.033 (0.945–1.128)	1.000 (0.854–1.170)	1.083 (0.939–1.249)
PM_{2.5} difference, µg/m³			
1-day	1.027 (0.986–1.070)	1.020 (0.952–1.092)	1.005 (0.938–1.077)
7-day	0.958 (0.904–1.014)	0.943 (0.854–1.042)	0.908 (0.828–0.996)
30-day	1.094 (0.990–1.210)	1.017 (0.859–1.204)	1.205 (1.011–1.436)

Table 4. Associations of the daily mean and difference in the ambient particulate matter with an aerodynamic diameter of $<2.5\ \mu\text{m}$ (PM_{2.5}), relative humidity (RH), and temperature for 1-, 7-, and 30-day averages with the step 5 treatments (triple inhaler, oral corticosteroids, and biologics). Data are presented as the odds ratio (OR) and 95% confident interval (CI). Abbreviations: OR, odds ratio; CI, confidence interval; particulate matter with an aerodynamic diameter of $<2.5\ \mu\text{m}$ (PM_{2.5}); relative humidity (RH). Covariates adjusted for age, body mass index (BMI), and smoking status. The bold value were indicates statistically significant ($p < 0.05$)

Because biologics are used in selected patients with severe asthma, this finding raises the possibility that fluctuating PM_{2.5} exposure may be particularly relevant in treatment-refractory

disease. PM_{2.5} can reach the distal airways, induce inflammation, impair lung function, and worsen symptom control.^{5,27} Exposure to PM_{2.5} has been associated with neutrophilic and

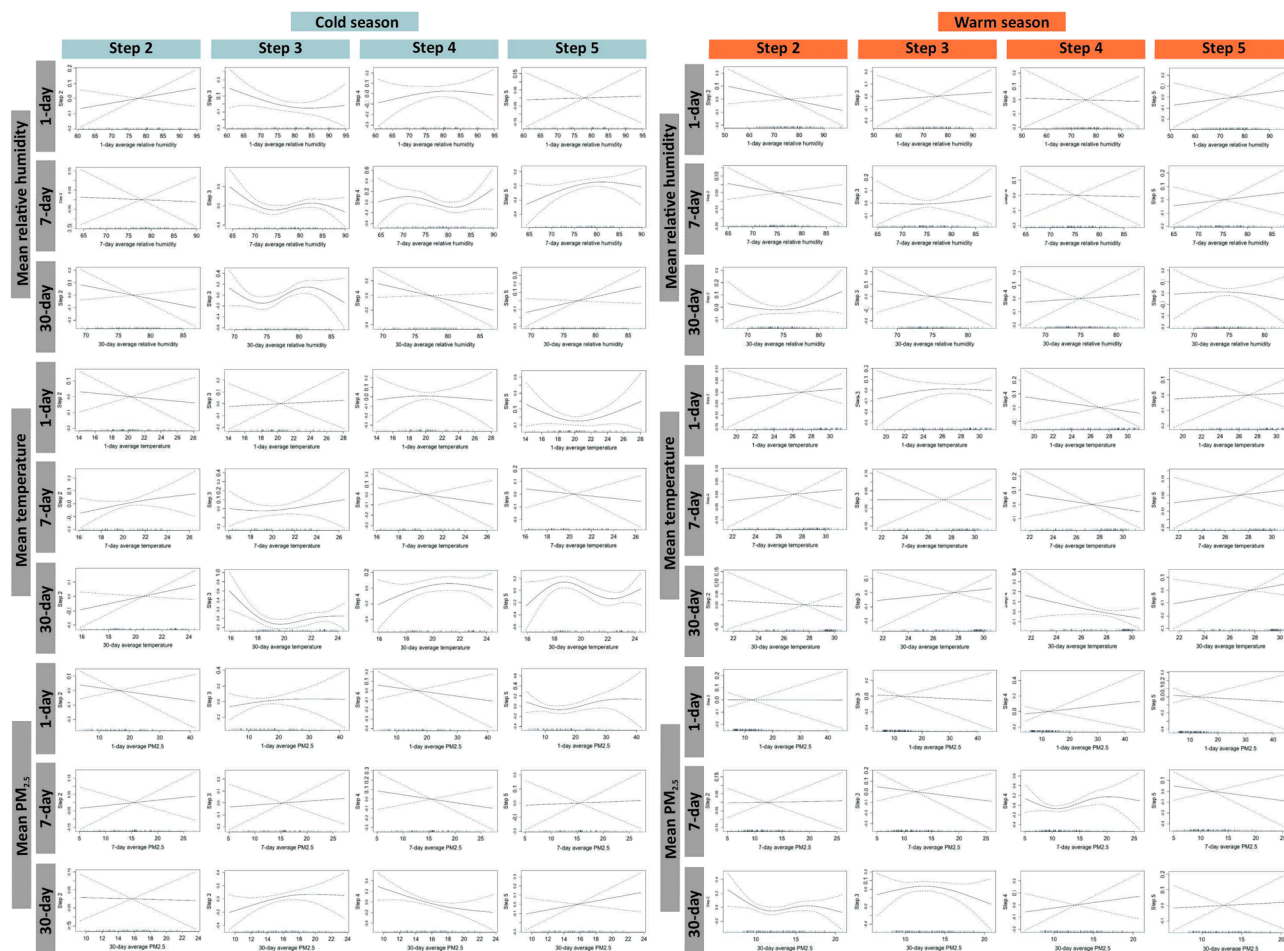


Fig. 1 Exposure-response relationship curves for the association of the daily average in the ambient relative humidity (RH) and temperature or particulate matter with an aerodynamic diameter of $<2.5 \mu\text{m}$ (PM_{2.5}) for 1-, 7-, and 30-day averages of cold and warm season with the asthma treatment (step 2, step 3, step 4, and step 5). Data are presented as the effects and 95% confident interval (CI). Covariates adjusted in the models were age, gender, body mass index (BMI), and smoking status.

eosinophilic inflammation, oxidative stress, and dysregulated type 1/type 2 immune responses.^{28,30} Prior population-based data have also shown that short-term PM_{2.5} exposure is associated with increased rescue inhaler use.²⁹ The biologic plausibility of an adverse PM_{2.5} effect is therefore supported. By contrast, the inverse association between 7-day PM_{2.5} difference and overall step 5 treatment was paradoxical and may reflect limited subgroup size, statistical instability, residual confounding, or multiple comparisons; accordingly, it should not be overinterpreted. Episodic PM_{2.5} exposure has been linked to increases in circulating inflammatory cells and proinflammatory mediators,³¹ which may partly explain why changes in exposure, rather than mean exposure alone, were associated with treatment intensity in our cohort.

The exposure-response analyses further suggested that treatment strategy was more sensitive to temperature and RH during the cold season. Previous studies have reported that cold seasons are associated with more severe respiratory symptoms, poorer asthma control, and increased medication use.³² We also observed an exposure-response relationship between PM_{2.5} and treatment strategy, although a clear seasonal pattern for PM_{2.5} was less evident. This is broadly consistent with prior studies linking indoor or ambient PM_{2.5} exposure to poorer asthma control in patients receiving ICS.³³ Taken together, our findings suggest that seasonal environmental instability, particularly during colder months, may increase the intensity of treatment required to achieve asthma control.

This study has several limitations. First, the study population was further divided into

treatment-step subgroups, which may have reduced statistical precision. Second, this was a single-center study of consecutively followed adult patients, which may limit external generalizability. Third, detailed information on indoor environmental exposure, occupational exposure, allergen burden, physical activity, and other air pollutants was not available. Fourth, although the lung function data were verified, the spirometric pattern in this real-world adult cohort may have been influenced by comorbidities and may not represent a classic obstructive phenotype in every patient. Fifth, multiple statistical comparisons were performed, and no formal adjustment for multiple testing was applied; therefore, some findings, particularly those with borderline statistical significance, should be interpreted with caution. Finally, because of the observational design, residual confounding cannot be excluded. Therefore, our findings should be interpreted as exploratory and hypothesis-generating and should be confirmed in larger multicenter studies.

CONCLUSION

In conclusion, greater short-term variation in RH and PM_{2.5} was associated with increased treatment intensity in adult asthma, especially during the cold season. RH variation was associated with step 4 treatment, mainly medium-dose ICS + LABA, whereas higher RH mean was associated with triple inhaler therapy and greater PM_{2.5} variation was associated with biologics use. These findings suggest that instability in meteorological and air pollution exposure may influence the treatment level required to achieve asthma control.

Abbreviations

BMI: body mass index; CWB: Central Weather Bureau; EPA: Environmental Protection Administration; ECP: eosinophil cationic protein; GINA: Global Initiative for Asthma; ICS: inhaled corticosteroids; LABA: long-acting beta-agonist; OR: odds ratio; OCS: oral corticosteroids; PM_{2.5}: particulate matter with an aerodynamic diameter of <2.5 µmR; OS: reactive oxygen species; RH: relative humidity; SABA: short-acting beta-agonist; SD: standard deviation.

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Data sharing

The data sets used and/or analyzed in this study are available from the corresponding author upon reasonable request.

Authors' contributions

FM, TYL, and HCC contributed to the completion of interpretation of the data and the manuscript. FM, YCL, TYL, and HCC contributed substantially to the concept, design, interpretation of the data, and completion of the study and manuscript. DR, N, and KFC contributed to critical revision of the manuscript for important intellectual content. SML, PJC and CYL contributed to the interpretation of the data and resources supporting. CHL, CHC, THC and PHT contributed to the interpretation of the data and project administration. KFC contributed to the data interpretation. All authors have read and approved the final manuscript.

Consent for publication

All authors reviewed the final version of the paper and agreed in writing to its publication in the World Allergy Organization Journal, as an open access article.

Ethics approval

All patients gave their written consent after being informed. This study was carried out in accordance with the protocol approved by the Chang Gung Medical Foundation Institutional Review Board (No.201900211B0).

Conflict of interest statement

No conflicts of interest to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.waojou.2026.101444>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT for English language revision to improve readability. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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